Week 10 Lab — Machine Learning & Dashboards

Train · Evaluate · Predict · Publish · Total: 35 marks (+3 bonus) · Due: end of Week 10

Name: _____ **Student ID:** _____ **Program:** _____

Instructions

• Work in a notebook named **Week10_YourName**. You need **week10_students.csv** (150 students, clean) next to the notebook.
• Use `test_size=0.25, random_state=42` in every split so your numbers match the checkpoints.
• **Submit:** (1) completed notebook, (2) `class_dashboard.html` + a screenshot of it open in your browser, (3) your **model card** (purpose, features, performance in plain language, risks), (4) screenshots.
• Colab: everything pre-installed. Anaconda: `pip install plotly` if the import fails.

The Dataset — week10_students.csv

150 students, clean (you earned it after Week 7). Columns: student_id · name · gender · program · district · attendance_pct · quiz_avg · assignments_avg · **score** (the target).

ID	Name	G	Program	District	Att%	Quiz	Assign	Score
AUCA001	Aline Uwase	F	Information Systems	Gasabo	55	31.0	35.0	28.6
AUCA002	Eric Iradukunda	M	Software Engineering	Musanze	73	61.1	76.7	64.5
AUCA003	Diane Uwizeyimana	F	Networking	Muhanga	92	35.2	65.5	52.7
AUCA004	Jean Bosco Ikirezi	M	Accounting	Kicukiro	69	65.2	54.2	68.3
AUCA005	Patrick Ingabire	M	Information Systems	Huye	88	39.3	42.9	42.3
...	... 145 more rows ...							

Part 1 · Explore & Split (5 marks)

Exercise 1.1 — Know your data (2 marks)

Load the CSV. Print the shape and the correlation of `score` with each of quiz_avg, assignments_avg, attendance_pct. In a comment: which feature looks most useful?

Exercise 1.2 — Frame & split (3 marks)

(a) Build `X` (three features) and `y` (score). (b) `train_test_split(X, y, test_size=0.25, random_state=42)`. (c) Print both set sizes and explain in a comment why 25% of students are hidden from the model.

Checkpoint: 112 train / 38 test.

Part 2 · Regression — Predict the Score (10 marks)

Exercise 2.1 — Train & read the model (3 marks)

Fit `LinearRegression` on the training data. Print each feature's coefficient with its name, plus the intercept. In a comment: one extra quiz point ≈ how many exam points?

Exercise 2.2 — Judge it honestly (4 marks)

Predict on the TEST set; print MAE and R^2. In a comment, translate both into plain language a dean would understand.

Checkpoint: MAE ≈ 3.3 points, R^2 ≈ 0.89.

Exercise 2.3 — Predict a new student (3 marks)

A new student: quiz 70, assignments 75, attendance 90 — build a one-row DataFrame (same column names) and predict their score. Then predict your own, with invented numbers.

Part 3 · Classification — Catch the At-Risk Students (8 marks)

Exercise 3.1 — Build the target & train (3 marks)

(a) `df["passed"] = (df["score"] >= 50).astype(int)`; print the pass/fail counts. (b) Split and train `LogisticRegression(max_iter=1000)`.

Exercise 3.2 — The accuracy trap (5 marks)

(a) Print the model's accuracy on the test set. (b) Print the lazy baseline — a model that predicts "pass" for everyone (`yc_test.mean()`). (c) Print the confusion matrix. (d) **In a comment (2 marks):** the two accuracies are nearly identical — is our model useless? Use the confusion matrix to argue.

Part 4 · Feature Engineering (5 marks)

Exercise 4.1 — Does more information help?

Train and evaluate (test-set R^2) three versions of the regression and print them side by side: (1) quiz_avg only, (2) all three numeric features, (3) all three + `program` one-hot encoded with `pd.get_dummies`. Same split settings every time. In a comment: what helped, what didn't, and what does that teach you?

Checkpoint: roughly 0.62 → 0.89 → 0.88.

Part 5 · Ship a Dashboard (7 marks)

Exercise 5.1 — The story in three charts (5 marks)

With Plotly Express, each with a title: (a) histogram of scores, 12 bins — "1. Where the class stands"; (b) scatter quiz_avg vs score with `color="passed"` and `hover_name="name"` — "2. Who is at risk"; (c) bar chart of average score per district — "3. Where to act".

Exercise 5.2 — Publish it (2 marks)

Save the at-risk scatter with `fig2.write_html("class_dashboard.html")`, open it in a browser, hover around, screenshot it for your submission.

Bonus (+3) — One chart, four stories

`px.box(df, x="program", y="score", color="program")` with a title. Two sentences: which program is most consistent, which has the widest spread?

Part 6 · The Model Card (required, checked for completion)

Write a short model card for your regression model: **1. Purpose** — what does it predict, for whom? **2. Data & features** — what went in? **3. Performance** — MAE and R^2 on unseen data, in plain language. **4. Risks** — who could be harmed by a wrong prediction, and how should the output be used (and NOT used)?

✓ Submission checklist

[ ] Notebook renamed Week10_YourName [ ] Regression evaluated on the TEST set [ ] Confusion-matrix argument written
[ ] class_dashboard.html saved + screenshot [ ] Model card written [ ] Submitted before end of Week 10

Marking scheme: Part 1 — 5 · Part 2 — 10 · Part 3 — 8 · Part 4 — 5 · Part 5 — 7 · Total 35 (+3 bonus).